NICOLAS SCHULZE

UNITY / C# DEVELOPER · SIMULATION, VR & GAMES

Based in Brazil (UTC–3) · Fully remote · Open to full-time or contract roles
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> **I build simulators where people learn to operate real heavy machinery, from vehicle physics and cab controls to the instructor's final grade report.**

KEY HIGHLIGHTS

CLIENTS	SENAI (Brazil's national industrial training service), Agropecuária Schio (a major apple producer) and an R&D project funded by Finep (Brazil's federal innovation agency)
SIMULATORS	5 training simulators across agriculture, industry, heavy equipment and VR on Meta Quest
GAMEPLAY	Vehicle and load physics, control remapping, step-based missions, HUD and gamepad-navigable menus
REAL HARDWARE	Steering wheels, pedals, joysticks and cab levers integrated with the software (Rewired)
4K PERFORMANCE	Zero per-frame memory allocations in steady state at 3840×2160 in the SENAI simulator
PIPELINE	Basic Blender and Substance for asset adjustments; UI/UX, physics and gameplay systems in Unity

SUMMARY

Unity and C# developer with nearly five years of experience building training simulators for agriculture, industry and heavy equipment, including VR. Takes systems from first version to long-term production maintenance: vehicle and implement physics, real cab controls, lesson flow and scoring, PDF reports, VR hand interaction and menus fully navigable by keyboard and gamepad.

Works in small teams of two to four developers, tracing root causes before fixing and validating every change. Familiar with the art pipeline for Unity; seeking roles in simulation and games, where physics, controls and UI make the difference.

EXPERIENCE

RDSIMTECH

Oct 2026 – Present

Software Development Engineer (Mid-level)

Continuation of the work started at RealDrive Tech: development and maintenance of Unity/C# training simulators, including the heavy-equipment simulator and the shared lesson package.

Heavy-Equipment Operator Training Simulator for SENAI (Yellow)

Jun 2026 – Present

Operator training for heavy machinery · core team of 4 developers · in progress

- **Lesson scoring:** built the evaluation for eight training programs, with one evaluator per program, score calculation and a PDF report with in-app preview and zoom
- **Real cab controls:** built on-screen guidance tied to the physical cab controls, so students know which lever or pedal to use, plus a control-remapping screen (Rewired Control Mapper) with pedal calibration
- **Mirrors:** built adjustable virtual and physical rear-view mirrors for each machine, saved per user (cameras and RenderTextures in HDRP)

- **Lesson lifecycle:** restart from the pause menu or the report without losing student and instructor data, and automatic failure on rollover
- **Performance & quality:** optimized lesson loading, memory allocations and mirror rendering, reaching zero per-frame allocations in steady state at 4K; wrote automated tests for reports, menus, hardware guidance and load retention

Shared Lesson & UI Package (RDLesson)

Sep 2025 — Present

Internal Unity package used by several of the company's simulators · team of 2

- Built a reusable UI kit (sidebar, cards, pop-ups) and keyboard and gamepad navigation across every menu
- Built report storage by student and lesson, with overwrite confirmation and automated tests
- Reworked the generic PDF report layout and added extension points so each simulator can customize its own
- **Result:** menus, navigation and reports built once and reused across several simulators

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REALDRIVE TECH

Jan 2022 — Oct 2026

Software Development Engineer (Mid-level)

Training simulators for industry, agriculture and heavy equipment, on multi-screen PC and in VR. Yellow and RDLesson started here and continue at RDSIMTECH.

Apple-Orchard & Warehouse Training Simulator (Schio)

Jul 2024 — May 2026

Client: Agropecuária Schio · one of 2 lead developers · delivered July 2025

- Worked on all eight training modules, including forklift, truck loading, orchard sprayer, trailer driving and the fruit-sizing machine
- Built student and instructor records with search and editing, training history and PDF report export
- Worked on penalty rules for collisions, rollovers and incorrect machine use, and built physics-based detection of dropped fruit bins
- Tuned trailer and implement physics (Rigidbody, joints), integrated the real tractor cab controls and helped migrate the project to HDRP
- **Result:** delivered to the client in July 2025 for operator training and maintained in production until May 2026

VR Industrial Training (RealDrive Industrial)

May 2025 — Apr 2026

Meta Quest 3 · Unity 6, URP, OpenXR, Meta XR SDK, XR Interaction Toolkit, XR Hands · main developer; sole developer on the overhead crane from Nov 2025

- **Industrial assembly (Cobra Correntes):** integrated the team's assembly systems and tools into the VR flow and produced content for 10+ industrial assemblies (chains, bearings, roller guides, clamps), with scenes, prefabs, work instructions and a technical clipboard
- **VR interaction & UI:** hand poses, grabbing without hand colliders pushing the object, two-handed scaling and World Space menus with a virtual keyboard (adapted MRTK), keeping the UI independent of the XR SDK
- **Lesson flow:** timer with pause, resume, time limit, automatic failure and report integration; safe save on exit or pause on Quest
- **Overhead crane:** extended a third-party crane package with a kinematic pivot and a ConfigurableJoint, so the load swings instead of jittering; built a step-based lesson manager with target zones and a timer; two lessons running in VR
- **Environment & tools:** three iterations of the industrial environment (URP, audio recorded at the real factory, 2D stand-ins to run on a standalone headset), a Unity Splines editor tool and 4K capture

Agricultural Machinery Simulator (RealDrive Agro)

Jan 2022 — Jun 2026

AGRO 4.0 project, funded by Finep · team of 3 · joined as a junior developer on the first version

- **Environment:** built two virtual farms with soybean, corn and wheat fields, using LODs and 2D impostors to render thousands of plants smoothly

- **Gameplay:** built the menu flow, a machine-selection showroom with color customization and the grain-unloading mechanic; integrated wheel and pedals through Rewired
- **Guided lessons:** built the steps of the planting, spraying and harvesting lessons and the machines' safety interlocks within the team's training manager
- **Telemetry & reporting:** built lesson data recording and, with another developer, an automatic PDF report with the machine's path, charts and statistics
- **Result:** showcased at Tecnoshow and Show Rural, two major Brazilian agricultural trade fairs

SKILLS

UNITY & C#	Unity 6, C#, HDRP and URP, editor tools, internal Unity packages, prefabs and serialization
PHYSICS & GAMEPLAY	Rigidbody, joints (ConfigurableJoint), vehicle and load physics, state machines, scoring and penalty rules
SIMULATION	Lesson flow, evaluation, telemetry, PDF reports (iTextSharp), training history and user records
HARDWARE INPUT	Steering wheels, pedals, joysticks and levers through Rewired and the Input System; control remapping and calibration
VIRTUAL REALITY	Meta Quest 3, Meta XR SDK, XR Interaction Toolkit, OpenXR, hand poses and two-handed interaction
UI/UX	Reusable components, menus and HUD, full keyboard and gamepad navigation, TextMeshPro
3D ART	Basic Blender and Substance, LODs and 2D impostors, environment composition and level design
QUALITY	Automated tests (edit mode and play mode), Profiler, loading time and memory allocations
WORKFLOW	Git, branching and code review in small teams; structured use of AI coding agents with human review of every change

PERSONAL PROJECT

House Farm · Environment & Capture Tool

2025

A furnished farmhouse interior built in Unity (HDRP), with environment composition and lighting, plus a custom C# tool to capture it in 4K images and multi-camera video directly from the editor.

EDUCATION

Self-taught · Unity, C# and Software Development

Learned on the job: joined as a junior developer on the first version of RealDrive Agro and learned Unity and C# directly in production projects, alongside independent study.

LANGUAGES

Portuguese (native) · English (elementary, A2 — technical reading)